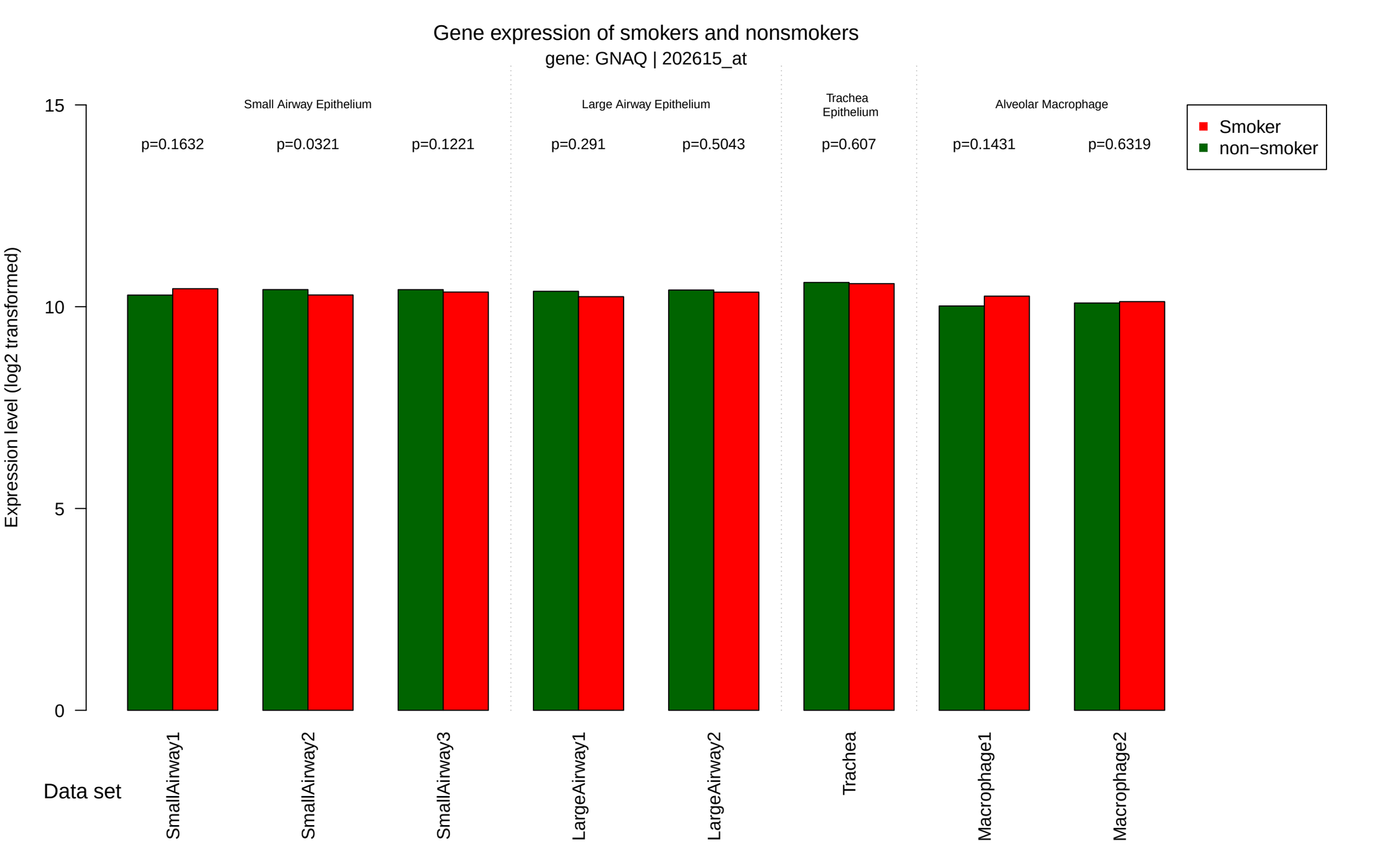
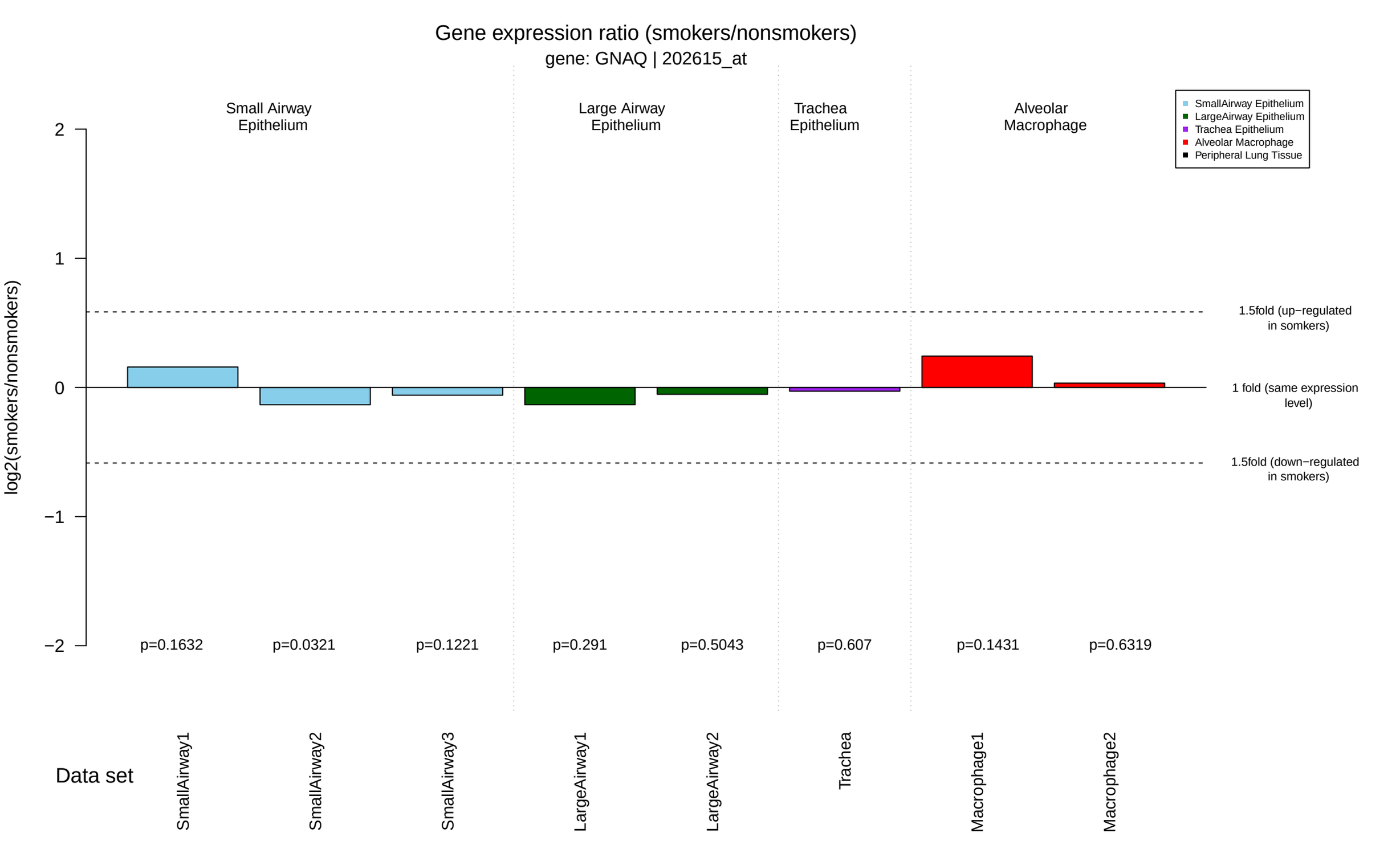
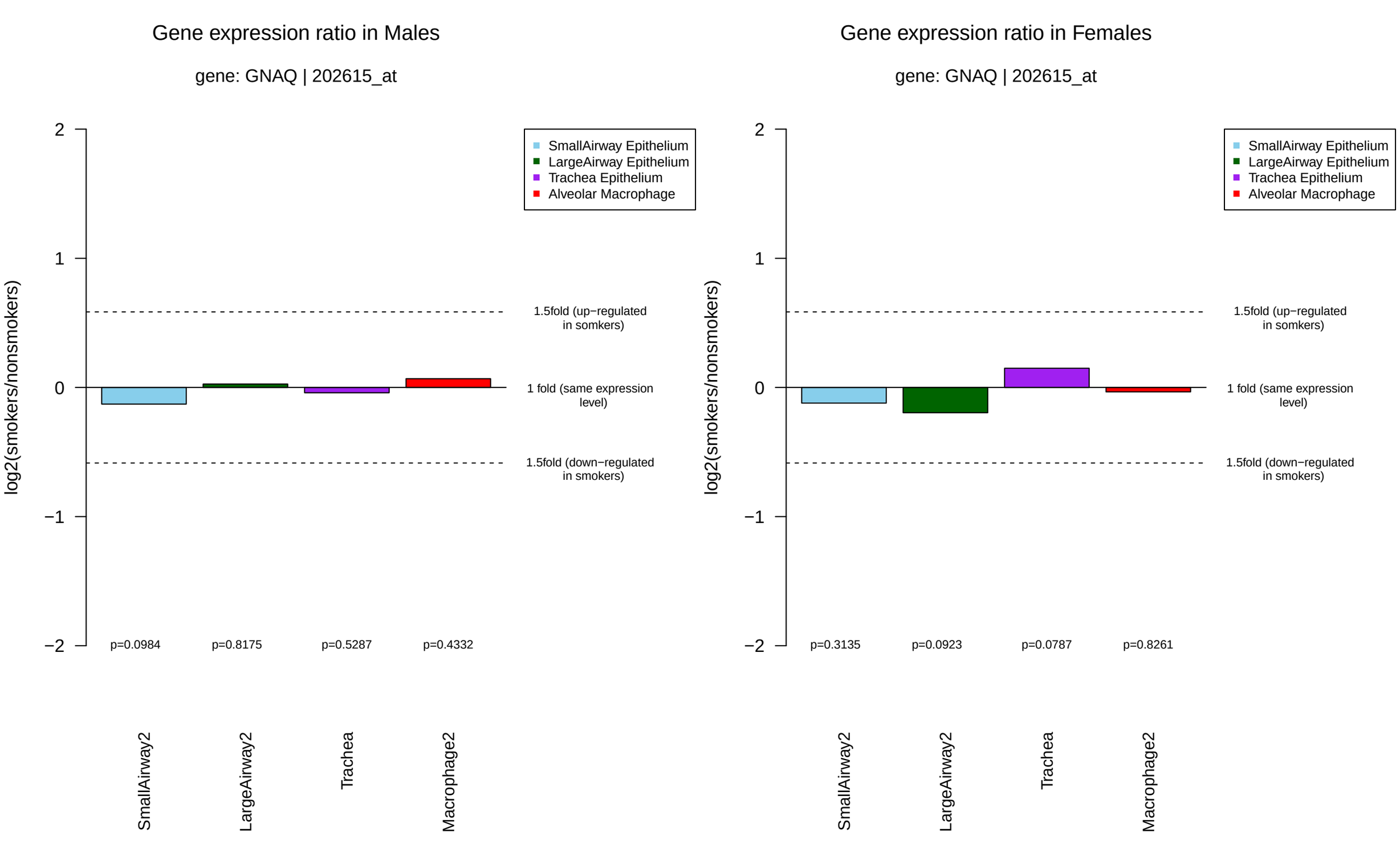
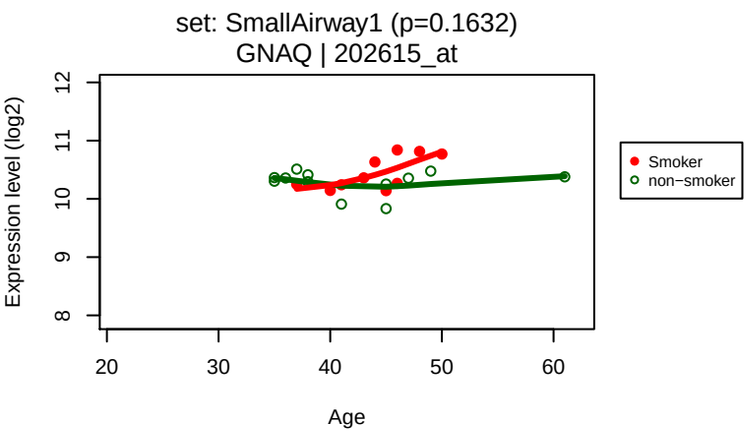
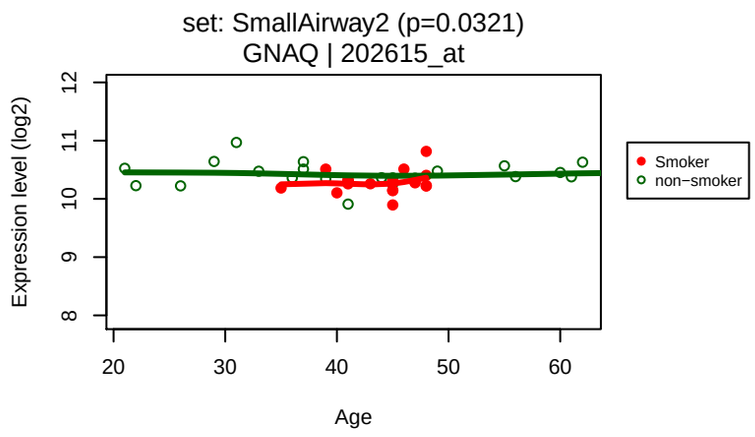
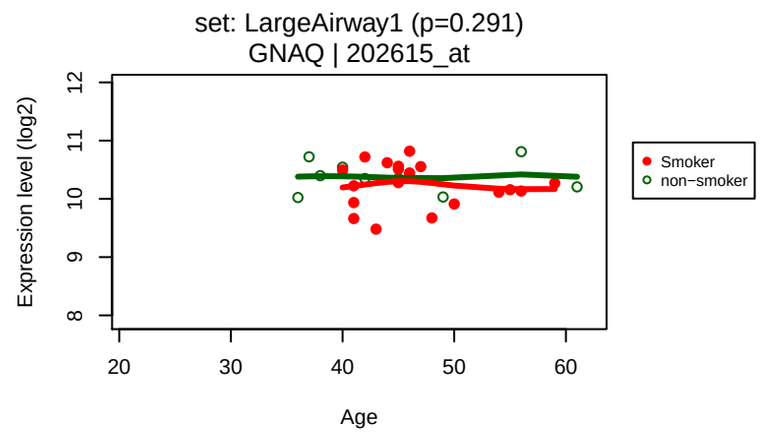

Gene expression of smokers and nonsmokers
gene: GNAQ | 202615_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1632
p=0.0321
p=0.1221
p=0.291
p=0.5043
p=0.607
p=0.1431
p=0.6319
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

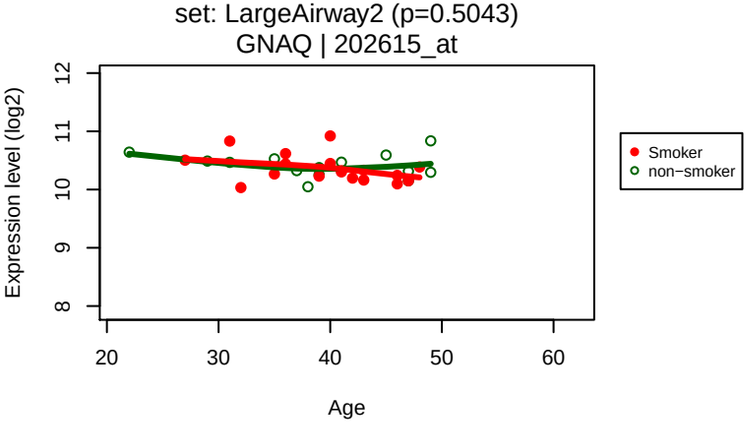

Gene expression ratio (smokers/nonsmokers)
gene: GNAQ | 202615_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1632
p=0.0321
p=0.1221
p=0.291
p=0.5043
p=0.607
p=0.1431
p=0.6319
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

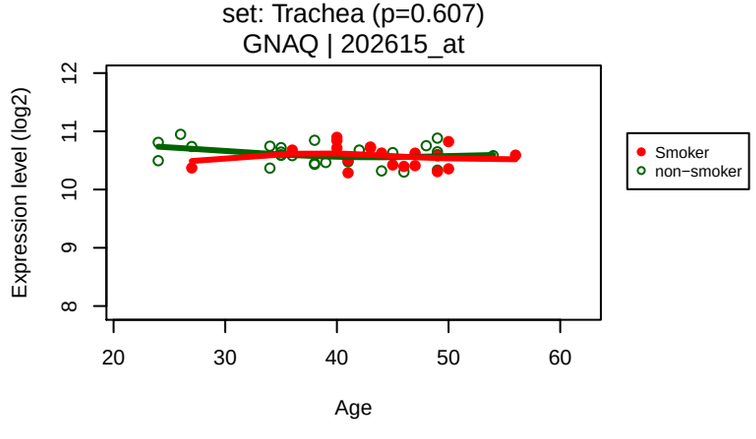

Gene expression ratio in Males
gene: GNAQ | 202615_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0984
p=0.8175
p=0.5287
p=0.4332
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNAQ | 202615_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3135
p=0.0923
p=0.0787
p=0.8261
SmallAirway2
LargeAirway2
Trachea
Macrophage2

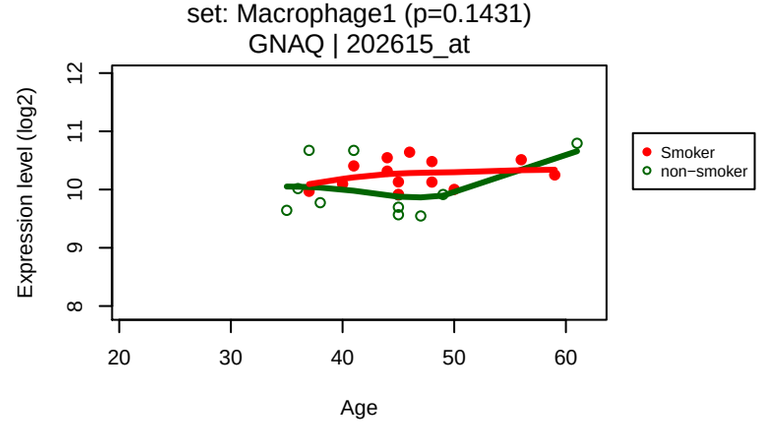

set: SmallAirway1 (p=0.1632)
GNAQ | 202615_at
Expression level (log2)
Age
Smoker
non−smoker

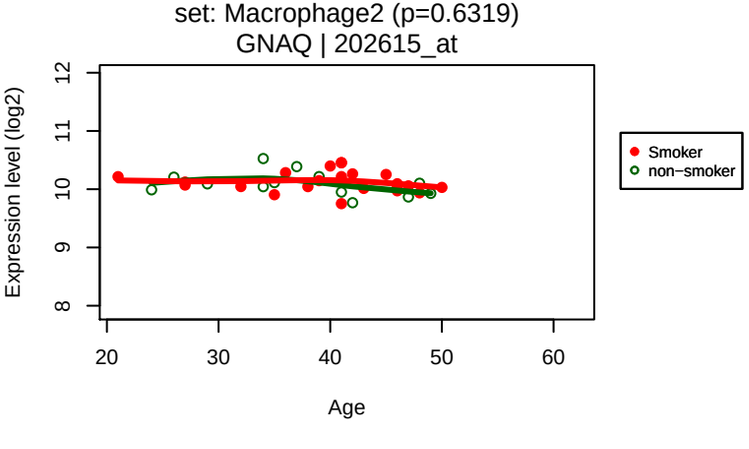

set: SmallAirway2 (p=0.0321)
GNAQ | 202615_at
Expression level (log2)
Age
Smoker
non−smoker

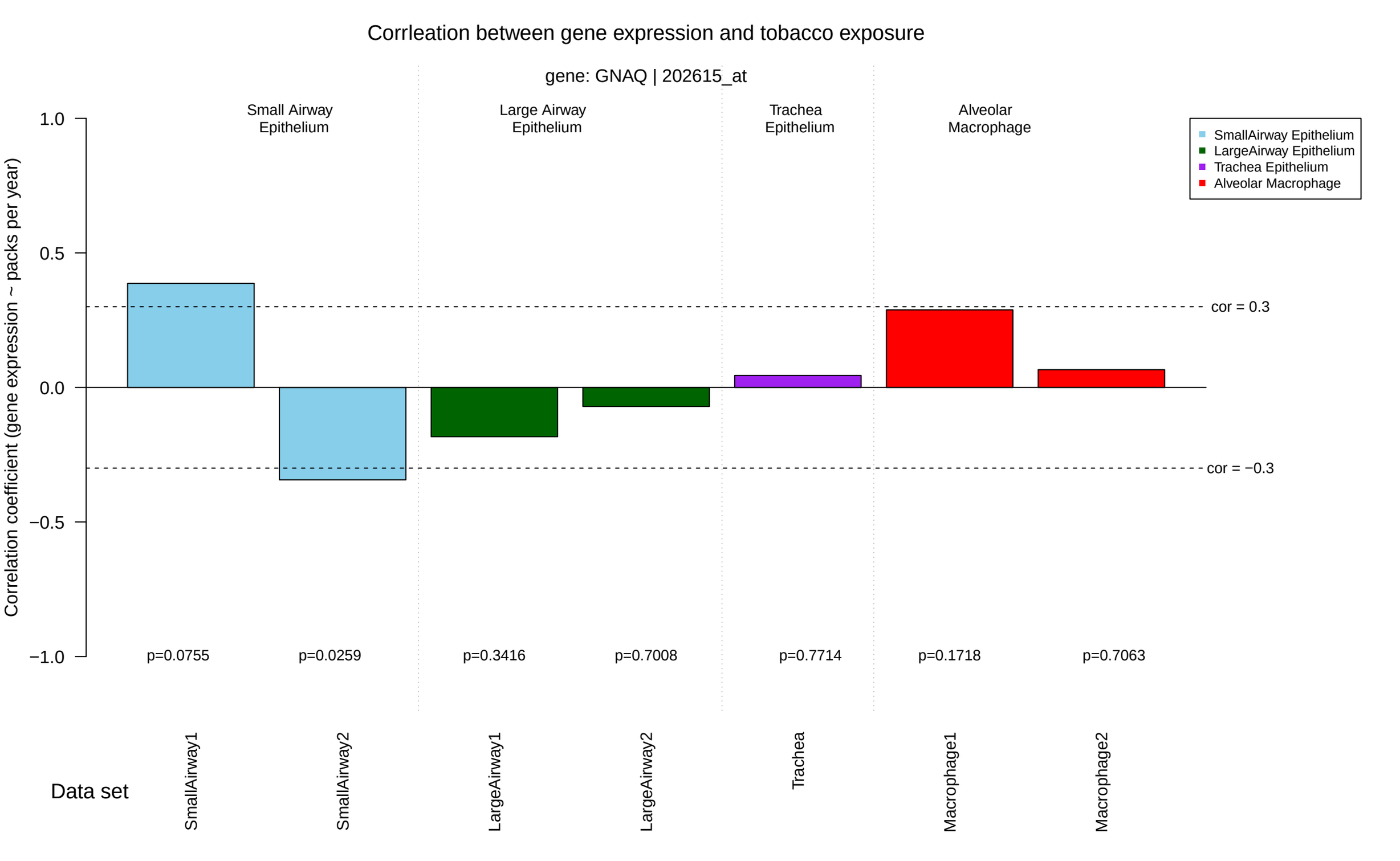

set: LargeAirway1 (p=0.291)
GNAQ | 202615_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5043)
GNAQ | 202615_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.607)
GNAQ | 202615_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1431)
GNAQ | 202615_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6319)
GNAQ | 202615_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNAQ | 202615_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0755
p=0.0259
p=0.3416
p=0.7008
p=0.7714
p=0.1718
p=0.7063
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2